

Hearsay

An educational speech-in-noise task with adaptive questions and a same-setup comparison board — UnivaBio 2026 · AI for Human Health · Sharon Basovich (solo)

Problem

Understanding speech in noise is often the first hearing ability to slip, and people often wait years before acting. Online hearing tests usually skip speech-in-noise, and they can't know how loud your headphones are — yet many still report a "normal" range.

Solution

- **Digits-in-noise check:** hear 3 spoken digits in speech-shaped noise, type them; antiphasic presentation; a headphone check first.
- **Bayesian adaptive engine** (the AI): posterior over threshold $\times$ slope; each next noise level minimizes expected posterior entropy; stops after at least 10 triplets when posterior SD is ≤ 1.25 dB, or at 24 triplets (11.8 on average in the simulated evaluation).
- **Same-device family board:** compares task results with the median of others on the same setup. A shared offset cancels under the model; real-world differences remain. The >3 dB flagging rule and comparisons are educational design choices, not clinically validated.
- **Hear-through simulator** and a reproducible in-browser **Lab**.

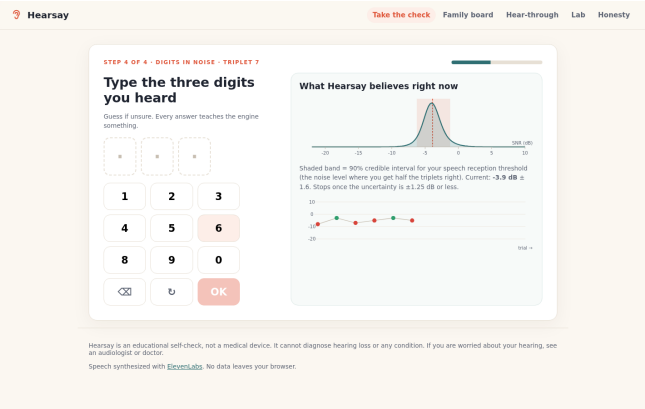
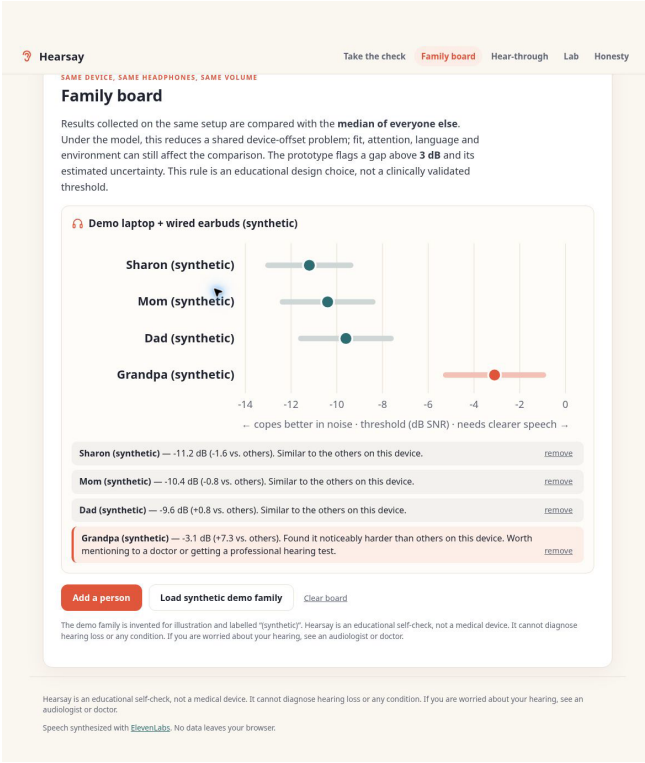
Audience & impact

Designed for families and adult learners exploring speech in noise. The intended value is making an adaptive listening task and its uncertainty easier to understand. Benefit to real users has not yet been evaluated.

Measured results (simulated listeners, n=400, tested twice)

Procedure	Triplets	RMSE	90% CI coverage
Staircase, 24 trials	24	0.79 dB	62%
Staircase, 12 trials	12	1.53 dB	54%
Hearsay Bayesian	11.8	1.01 dB	94%

~34% lower error than a staircase of the same length; half the trials of the classic track, which remains slightly more precise.



Built

TypeScript, Vite, Web Audio API, hand-written FFT, Vitest (32 tests). No backend; results stay in the browser.

Honesty

Not a medical device; not a diagnosis; not clinically validated. Validation uses simulated listeners only. Demo family is invented and labelled. Speech synthesized once, offline, with ElevenLabs TTS; noise generated locally; live site calls no APIs. Code, tests and docs generated with the Devin AI coding agent. No human-subject data. Planned next: ethics-approved pilot against a calibrated clinical test.